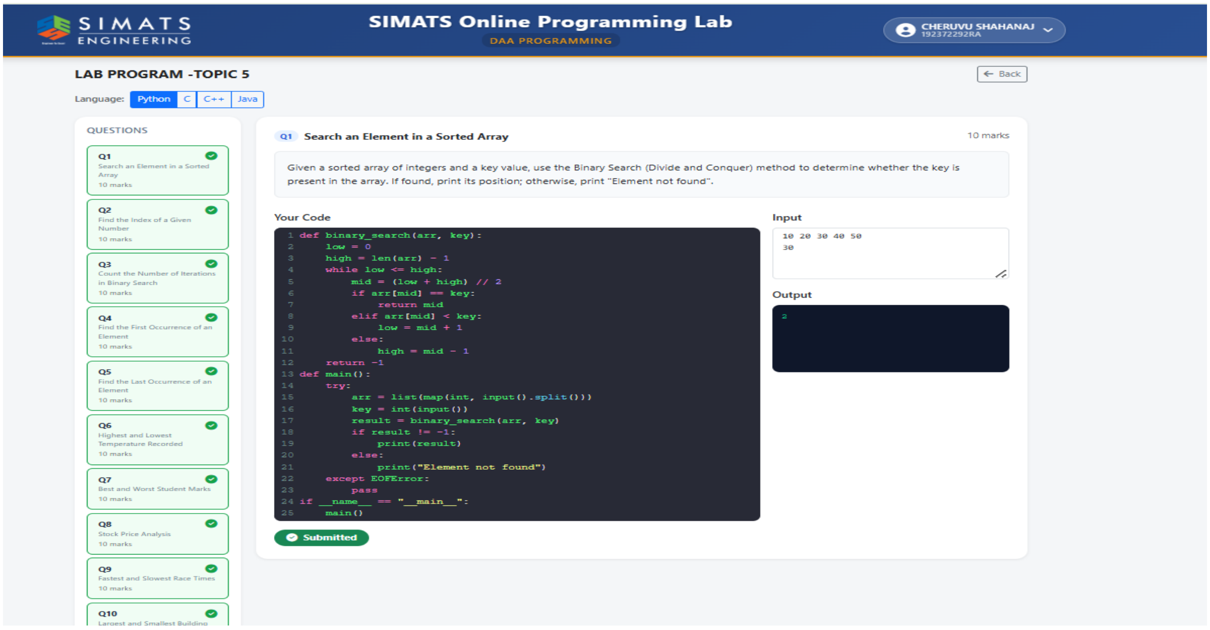
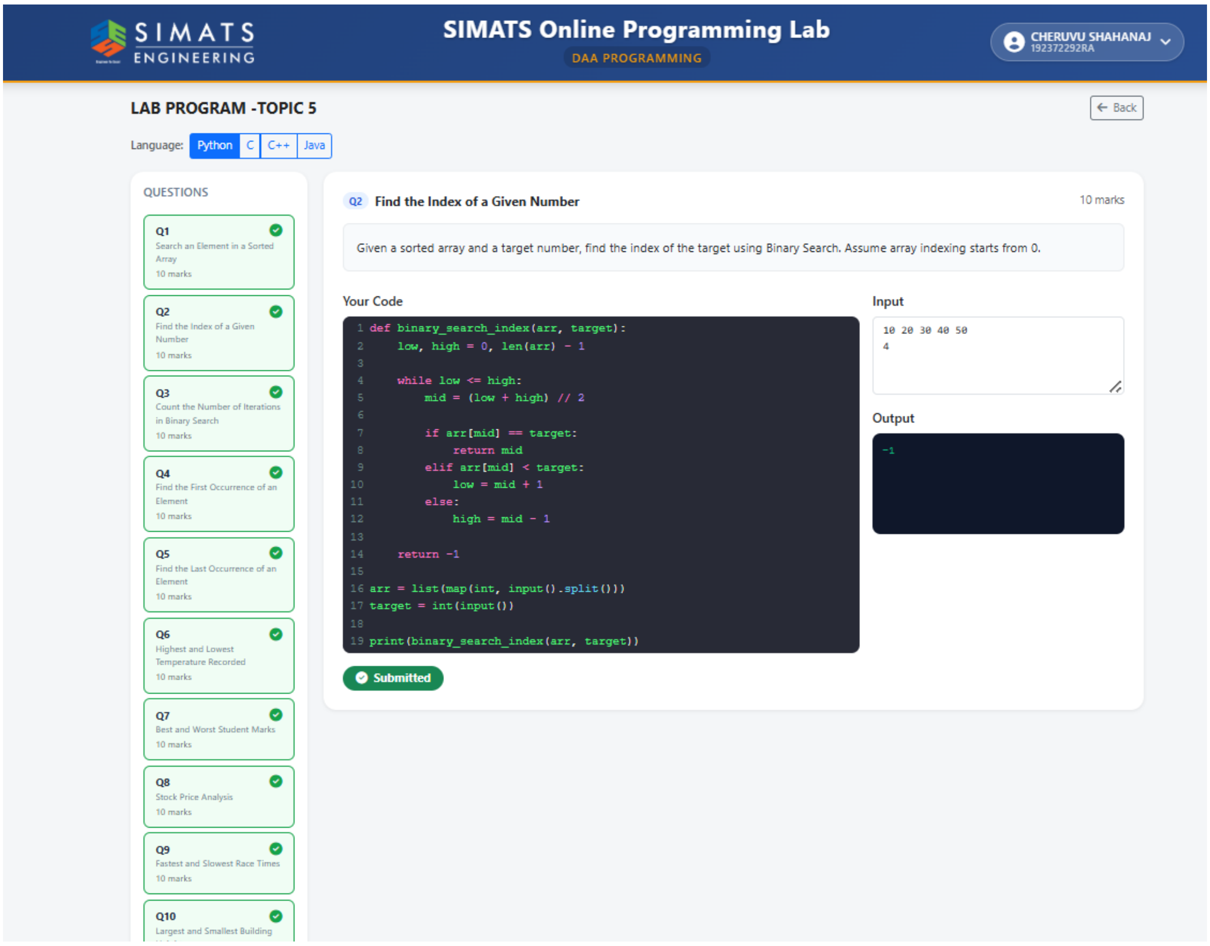
LAB PROGRAM – TOPIC 5

SHAHANAJ C

192372292





SIMATS ENGINEERING

SIMATS Online Programming Lab

DAA PROGRAMMING

CHERUVU SHAHANAJ 192372292RA

LAB PROGRAM -TOPIC 5

Back

Language: Python | C | C++ | Java

QUESTIONS

- Q1 Search an Element in a Sorted Array 10 marks
- Q2 Find the Index of a Given Number 10 marks
- Q3 Count the Number of Iterations in Binary Search 10 marks
- Q4 Find the First Occurrence of an Element 10 marks
- Q5 Find the Last Occurrence of an Element 10 marks
- Q6 Highest and Lowest Temperature Recorded 10 marks
- Q7 Best and Worst Student Marks 10 marks
- Q8 Stock Price Analysis 10 marks
- Q9 Fastest and Slowest Race Times 10 marks
- Q10 Largest and Smallest Building Heights

Q3 Count the Number of Iterations in Binary Search 10 marks

Given a sorted array and a key, perform Binary Search and display the number of iterations required to find the element.

Your Code

```python
def binary_search_with_iterations(arr, key):
    low = 0
    high = len(arr) - 1
    iterations = 0

    while low <= high:
        iterations += 1
        mid = (low + high) // 2

        if arr[mid] == key:
            return mid, iterations
        elif arr[mid] < key:
            low = mid + 1
        else:
            high = mid - 1

    return -1, iterations


try:

    arr = list(map(int, input().split()))

    key = int(input())

    index, count = binary_search_with_iterations(arr, key)

    if index != -1:
        print(f"Element found at index: {index}")
    else:
        print("Element not found")

    print(f"Number of iterations: {count}")

except EOFError:
    pass
```

Input

```
2 3 4 5 6 7 8 9
4
```

Output

```
Element found at index: 2
Number of iterations: 3
```

SIMATS ENGINEERING

SIMATS Online Programming Lab

DAA PROGRAMMING

CHERUVU SHAHANAJ 192372292RA

LAB PROGRAM -TOPIC 5

Back

Language: Python | C | C++ | Java

QUESTIONS

- Q1 Search an Element in a Sorted Array 10 marks
- Q2 Find the Index of a Given Number 10 marks
- Q3 Count the Number of Iterations in Binary Search 10 marks
- Q4 Find the First Occurrence of an Element 10 marks
- Q5 Find the Last Occurrence of an Element 10 marks
- Q6 Highest and Lowest Temperature Recorded 10 marks
- Q7 Best and Worst Student Marks 10 marks
- Q8 Stock Price Analysis 10 marks
- Q9 Fastest and Slowest Race Times 10 marks
- Q10 Largest and Smallest Building

Q4 Find the First Occurrence of an Element 10 marks

Given a sorted array containing duplicate elements, find the first occurrence of a given key using Binary Search.

Your Code

```python
def find_first_occurrence(arr, key):
    low = 0
    high = len(arr) - 1
    result = -1

    while low <= high:
        mid = (low + high) // 2

        if arr[mid] == key:
            result = mid
            high = mid - 1
        elif arr[mid] < key:
            low = mid + 1
        else:
            high = mid - 1

    return result


if __name__ == "__main__":
    try:

        data = input().split()
        if not data:
            arr = []
        else:
            arr = [int(x) for x in data]

        key = int(input())

        print(find_first_occurrence(arr, key))
    except EOFError:
        pass
```

Input

```
12 23 45 67 12 5 6
5
```

Output

```
-1
```

Submitted

SIMATS ENGINEERING

SIMATS Online Programming Lab

DAA PROGRAMMING

CHERUVU SHAHANAJ
192372292RA

LAB PROGRAM -TOPIC 5

Back

Language: Python | C | C++ | Java

QUESTIONS

- **Q1** Search an Element in a Sorted Array — 10 marks
- **Q2** Find the Index of a Given Number — 10 marks
- **Q3** Count the Number of Iterations in Binary Search — 10 marks
- **Q4** Find the First Occurrence of an Element — 10 marks
- **Q5** Find the Last Occurrence of an Element — 10 marks
- **Q6** Highest and Lowest Temperature Recorded — 10 marks
- **Q7** Best and Worst Student Marks — 10 marks
- **Q8** Stock Price Analysis

Q5 Find the Last Occurrence of an Element

10 marks

Given a sorted array containing duplicate elements, find the last occurrence of a given key using Binary Search.

Your Code

```python
def last_occurrence(arr, key):
    low, high = 0, len(arr) - 1
    ans = -1

    while low <= high:
        mid = (low + high) // 2

        if arr[mid] == key:
            ans = mid
            low = mid + 1
        elif arr[mid] < key:
            low = mid + 1
        else:
            high = mid - 1

    return ans
```

Input

```
1 2 2 2 3 4
2
```

Output

```
3
```

Submitted

SIMATS ENGINEERING

SIMATS Online Programming Lab

DAA PROGRAMMING

CHERUVU SHAHANAJ
192372292RA

LAB PROGRAM -TOPIC 5

Back

Language: Python | C | C++ | Java

QUESTIONS

- **Q1** Search an Element in a Sorted Array — 10 marks
- **Q2** Find the Index of a Given Number — 10 marks
- **Q3** Count the Number of Iterations in Binary Search — 10 marks
- **Q4** Find the First Occurrence of an Element — 10 marks
- **Q5** Find the Last Occurrence of an Element — 10 marks
- **Q6** Highest and Lowest Temperature Recorded — 10 marks
- **Q7** Best and Worst Student Marks — 10 marks
- **Q8** Stock Price Analysis — 10 marks
- **Q9** Fastest and Slowest Race Times — 10 marks
- **Q10**

Q6 Highest and Lowest Temperature Recorded

10 marks

A weather station records temperatures over a week. Find the minimum and maximum temperature using the Divide and Conquer approach.

Your Code

```python
def find_min_max(arr, low, high):
    if low == high:
        return arr[low], arr[low]
    if high == low + 1:
        if arr[low] < arr[high]:
            return arr[low], arr[high]
        else:
            return arr[high], arr[low]
    mid = (low + high) // 2
    left_min, left_max = find_min_max(arr, low, mid)
    right_min, right_max = find_min_max(arr, mid + 1, high)
    overall_min = min(left_min, right_min)
    overall_max = max(left_max, right_max)
    return overall_min, overall_max
def main():
    try:
        input_data = input().split()
        if not input_data:
            return
        temperatures = [float(x) for x in input_data]

        if len(temperatures) == 0:
            return
        min_temp, max_temp = find_min_max(temperatures, 0, len(temperat
        print(f"Minimum Temperature: {min_temp}")
        print(f"Maximum Temperature: {max_temp}")

    except EOFError:
        pass
    except ValueError:
        pass

if __name__ == "__main__":
    main()
```

Input

```
23 25.5 19 28 21 24
```

Output

```
Minimum Temperature: 19.0
Maximum Temperature: 28.0
```

SIMATS ENGINEERING

SIMATS Online Programming Lab

DAA PROGRAMMING

CHERUVU SHAHANAJ 192372292RA

LAB PROGRAM -TOPIC 5

Back

Language: Python | C | C++ | Java

QUESTIONS

- Q1 Search an Element in a Sorted Array 10 marks
- Q2 Find the Index of a Given Number 10 marks
- Q3 Count the Number of Iterations in Binary Search 10 marks
- Q4 Find the First Occurrence of an Element 10 marks
- Q5 Find the Last Occurrence of an Element 10 marks
- Q6 Highest and Lowest Temperature Recorded 10 marks
- Q7 Best and Worst Student Marks 10 marks
- Q8 Stock Price Analysis 10 marks
- Q9 Fastest and Slowest Race Times 10 marks
- Q10 Largest and Smallest Building Heights

Q7 Best and Worst Student Marks 10 marks

A teacher wants to identify the highest and lowest marks scored in a class using Divide and Conquer.

Your Code

```python
def find_max_min(arr, low, high):
    if low == high:
        return arr[low], arr[low]
    if high == low + 1:
        if arr[low] < arr[high]:
            return arr[high], arr[low]
        else:
            return arr[low], arr[high]
    mid = (low + high) // 2
    max1, min1 = find_max_min(arr, low, mid)
    max2, min2 = find_max_min(arr, mid + 1, high)
    overall_max = max(max1, max2)
    overall_min = min(min1, min2)

    return overall_max, overall_min

def main():
    try:
        data = input().split()
        if not data:
            return

        marks = [int(x) for x in data]

        if len(marks) == 0:
            return

        max_val, min_val = find_max_min(marks, 0, len(marks) - 1)
        print(f"Highest: {max_val}")
        print(f"Lowest: {min_val}")
    except EOFError:
        pass

if __name__ == "__main__":
    main()
```

Input

```
10 20 5 30 25
```

Output

```
Highest: 30
Lowest: 5
```

SIMATS ENGINEERING

SIMATS Online Programming Lab

DAA PROGRAMMING

CHERUVU SHAHANAJ 192372292RA

LAB PROGRAM -TOPIC 5

Back

Language: Python | C | C++ | Java

QUESTIONS

- Q1 Search an Element in a Sorted Array 10 marks
- Q2 Find the Index of a Given Number 10 marks
- Q3 Count the Number of Iterations in Binary Search 10 marks
- Q4 Find the First Occurrence of an Element 10 marks
- Q5 Find the Last Occurrence of an Element 10 marks
- Q6 Highest and Lowest Temperature Recorded 10 marks
- Q7 Best and Worst Student Marks 10 marks
- Q8 Stock Price Analysis 10 marks

Q8 Stock Price Analysis 10 marks

An investor wants to know the minimum and maximum stock prices recorded during a trading week.

Your Code

```python
def stock_min_max(prices):
    if not prices:
        return None, None

    min_price = prices[0]
    max_price = prices[0]

    for p in prices[1:]:
        if p < min_price:
            min_price = p
        if p > max_price:
            max_price = p

    return min_price, max_price
prices = list(map(int, input().split()))
min_price, max_price = stock_min_max(prices)

print(min_price, max_price)
```

Input

```
10 12 7 17 9
```

Output

```
7 17
```

Submitted

SIMATS ENGINEERING

SIMATS Online Programming Lab

DAA PROGRAMMING

CHERUVU SHAHANAJ 192372292RA

LAB PROGRAM -TOPIC 5

Back

Language: Python | C | C++ | Java

QUESTIONS

- Q1 Search an Element in a Sorted Array — 10 marks
- Q2 Find the Index of a Given Number — 10 marks
- Q3 Count the Number of Iterations in Binary Search — 10 marks
- Q4 Find the First Occurrence of an Element — 10 marks
- Q5 Find the Last Occurrence of an Element — 10 marks
- Q6 Highest and Lowest Temperature Recorded — 10 marks
- Q7 Best and Worst Student Marks — 10 marks
- Q8 Stock Price Analysis — 10 marks
- Q9 Fastest and Slowest Race Times — 10 marks
- Q10

Q9 Fastest and Slowest Race Times

10 marks

A sports analyst needs to find the fastest (minimum) and slowest (maximum) race completion times.

Your Code

```python
def find_fastest_and_slowest(times):
    if not times:
        return None, None

    fastest = min(times)
    slowest = max(times)

    return fastest, slowest

if __name__ == "__main__":)
    try:

        user_input = input()
        race_times = [float(t) for t in user_input.split()]

        if race_times:
            fastest, slowest = find_fastest_and_slowest(race_times)
            print(f"{fastest} {slowest}")
        else:
            print("No times provided")
    except ValueError:
        print("Invalid input. Please enter numerical values.")
```

Submitted

Input

```
12.5 10.2 15.8 9.8 14.1
```

Output

```
9.8 15.8
```

SIMATS ENGINEERING

SIMATS Online Programming Lab

DAA PROGRAMMING

CHERUVU SHAHANAJ 192372292RA

LAB PROGRAM -TOPIC 5

Back

Language: Python | C | C++ | Java

QUESTIONS

- Q1 Search an Element in a Sorted Array — 10 marks
- Q2 Find the Index of a Given Number — 10 marks
- Q3 Count the Number of Iterations in Binary Search — 10 marks
- Q4 Find the First Occurrence of an Element — 10 marks
- Q5 Find the Last Occurrence of an Element — 10 marks
- Q6 Highest and Lowest Temperature Recorded — 10 marks
- Q7 Best and Worst Student Marks — 10 marks
- Q8 Stock Price Analysis — 10 marks
- Q9 Fastest and Slowest Race Times — 10 marks

Q10 Largest and Smallest Building Heights

10 marks

An urban planner is analyzing building heights in a city block and wants to determine the tallest and shortest buildings.

Your Code

```python
def find_heights():
    try:
        heights = list(map(int, input().split()))

        if not heights:
            return
        tallest = max(heights)
        shortest = min(heights)
        print(f"Tallest Building: {tallest}")
        print(f"Shortest Building: {shortest}")

    except ValueError:
        print("Please enter valid integer heights.")

if __name__ == "__main__":
    find_heights()
```

Submitted

Input

```
5
12 5 19 3 10
```

Output

```
Tallest Building: 5
Shortest Building: 5
```